

Autonomous Navigation in Fire Environment Using Drone

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Abstract—Autonomous navigation in indoor fire-affected environments poses significant challenges due to extreme temperatures, dense smoke, and highly nonlinear thermal behavior. Lightweight micro-UAVs such as the Crazyflie 2.1 offer a safe alternative for fire exploration, provided they can interpret local temperature cues and make reliable navigation decisions. This work presents a reinforcement-learning-based navigation framework integrated with the Webots–ROS ecosystem, enabling a micro-drone to locate high-temperature regions using only local thermal measurements. A dual temperature-generation pipeline—comprising physics-based Pyrosim simulations and analytical temperature fields—provides diverse fire scenarios for evaluation. Three decision-making strategies are implemented and compared: greedy hill climbing, ϵ -greedy multi-armed bandits, and Upper-Confidence-Bound (UCB) bandits. Numerical experiments across single-source, multi-source, and sink-distorted temperature fields demonstrate that bandit-based approaches achieve more robust performance in flat-gradient or misleading thermal landscapes, while greedy hill climbing excels in well-behaved fields. The results highlight the feasibility of temperature-driven autonomous navigation for micro-UAVs operating in hazardous indoor fire environments.

Index Terms—ROS, Webot, Pyrosim, Exploration, Exploitation, Greedy, ϵ -Greedy, UCB

I. INTRODUCTION

Autonomous navigation in indoor fire-affected environments is a challenging problem due to extreme temperatures, close environment, and the risks posed to human responders. Small aerial robots such as Crazyflie 2.1 offer a promising alternative to safe exploration and monitoring of hazardous environments, but their effectiveness is highly dependent on their ability to perceive and react to thermal conditions in real time. Traditional vision- or LiDAR-based navigation becomes unrealistic due to the small size of drones as well as unreliability in smoke-filled or low-visibility fire scenarios, making temperature-driven navigation an attractive alternative. To address this challenge, this work develops a reinforcement learning-based navigation framework in which the drone uses local temperature measurements to infer optimal movement strategies within an indoor fire environment. By leveraging both physics-based fire simulations and mathematically generated temperature fields, the proposed system aims to evaluate how lightweight micro-drones can autonomously locate and navigate towards fire sources, supporting future applications in fire assessment, search-and-rescue assistance, and hazardous-

environment monitoring. Although in this paper, the RL-ROS-Webot framework has been used for fire source detection, it can also be used for localization of other sources, viz. Chemical or gas sources based on their spatial-temporal distributions.

Recent research on fire-related sensing and UAV navigation spans experimental fire-temperature studies, multi-UAV coordination strategies, and reinforcement-learning-based exploration. Experimental works have analyzed smoke and temperature distributions in indoor multi-room environments, providing valuable spatial thermal-flow insights but focusing solely on passive measurement rather than active navigation [1]. UAV-based firefighting and monitoring approaches have explored dynamic multi-swarm coordination for large-scale outdoor fires, emphasizing coverage, communication, and collective response [2]. Reinforcement-learning frameworks have also been applied to forest-fire localization, primarily targeting fire-region discovery and propagation modeling rather than precise navigation within strongly varying temperature fields [3].

Now, coming to the problem of navigating a micro-drone through fire-affected environments, the framework we developed consists of a reinforcement-learning-based navigation algorithm integrated with the Webot simulation platform and ROS middle-ware [4]. The system uses a mathematically defined temperature field as the primary environmental cause, and the RL agent selects target positions based on the temperature readings at the current location of the drone. A proportional controller then converts the target position into velocity commands, allowing the drone to move through the environment in a closed-loop manner. The framework is evaluated in multiple temperature distributions to assess its robustness under varying fire conditions.

The main contributions of this work are summarized as follows:

- We develop a reinforcement learning-based navigation framework that enables a Crazyflie 2.1 micro-quadrotor to autonomously navigate toward high-temperature regions in simulated fire environments.
- We integrate the framework with the Webot–ROS ecosystem, enabling closed-loop interaction between the simulated environment, the RL policy, and the drone controller.

- We construct a dual fire–temperature generation pipeline: (i) a physics-based Pyrosim setup for realistic fire simulations and (ii) computationally efficient analytical temperature fields that support rapid experimentation while preserving realistic spatial characteristics.
- We evaluate the proposed framework in multiple fire scenarios, demonstrating its ability to navigate effectively in diverse thermal distributions, and compare the effectiveness of different RL algorithms.

II. RELATED WORK

Recent research has extensively explored fire temperature modeling and UAV-based fire exploration. Full-scale experimental studies have examined indoor fire behavior, revealing complex spatial distributions of smoke and temperature in enclosed and multi-room environments [1]. Other works have investigated UAV-based strategies for wildfire monitoring and localization. Swarm-based approaches employ coordinated multi-UAV systems for fire detection and suppression [2], while reinforcement-learning-driven UAV exploration has been proposed for identifying and tracking fire propagation in outdoor environments [3].

Despite these advancements, several limitations persist, especially in closed or indoor spaces. Experimental studies provide passive thermal measurements but do not couple these observations with active robot navigation or decision-making frameworks. UAV-based RL approaches primarily focus on fire-front tracking or large-scale propagation modeling rather than fine-grained navigation within varying thermal fields. Furthermore, many existing methods assume coarse or low-resolution thermal inputs, which limits their applicability to dense indoor temperature gradients.

Our work addresses these gaps by enabling a micro-UAV to autonomously navigate toward thermal peaks within simulated indoor-like fire environments. Unlike prior approaches, we emphasize fine-resolution temperature-driven navigation and evaluate performance across multiple controlled fire scenarios. This allows us to study navigation behavior directly in relation to thermal gradients, filling the gap between detailed fire-temperature modeling and active UAV decision-making in confined environments.

III. METHODOLOGY

A. System Overview

The objective of the proposed system is to enable autonomous navigation of a Crazyflie 2.1 micro–quadrotor in simulated fire environments. The drone obtains the temperature at its current location from the simulated environment and uses this information as input for a reinforcement learning (RL) framework. Based on the temperature value received, the RL agent predicts the target position, and a proportional controller computes velocity commands by evaluating the error between the current and target positions. The loop continues until the error falls below a predefined threshold, indicating that the drone has reached the desired location.

The robotic platform used in this work is the Crazyflie 2.1 model available in the Webot simulation environment. All sensing is simulated, and the temperature field is generated mathematically using predefined temperature distribution equations. Given the robot’s (x, y) coordinates, the simulator computes the corresponding temperature by directly evaluating these equations. Multiple fire scenarios are generated with varying spatial distributions of temperature to evaluate the navigation capability of the proposed framework.

The system architecture consists of four major modules: the fire simulation module, the RL agent, the Webots simulation environment, and the ROS middleware. Communication between modules is handled through ROS topics, while Python scripts implement the RL algorithm and the Crazyflie controller. The RL node operates as a ROS node written in Python, and closed-loop control is maintained throughout the simulation. The overall system follows a feedback interaction cycle: the fire simulation provides temperature information, the RL agent selects an action, velocity commands are generated by the controller, the drone moves within Webots, and the updated state is returned to the RL node.

This work assumes that the fire is static during each episode, under the justification that although the magnitude of the temperature field may change over time, the global maximum location remains constant. It is further assumed that the temperature at the robot’s position is known exactly. The experiments are performed in a 2D environment, and a heatmap of the temperature distribution with the drone’s trajectory is used for visualization in the results.

B. Fire Simulation

Fire behavior in indoor environments is highly nonlinear and arises from complex fluid–thermal interactions, making realistic temperature modeling essential for evaluating autonomous navigation. To incorporate such conditions, Pyrosim is employed as it provides physics-based fire simulations using the Fire Dynamics Simulator (FDS) backend, offering spatially resolved temperature fields that integrate into a ROS-based pipeline. The primary motivation for using Pyrosim lies in its ability to generate realistic, customizable fire temperature distributions that closely resemble physical fire dynamics [5]. In this work, Pyrosim is used to validate the integration of real fire-simulation output with the proposed RL-driven navigation framework through a simple single-source fire scenario, where the temperature field exhibits circular symmetry with a maximum at the fire center. This configuration serves as a controlled baseline for assessing the performance of the temperature-dependent navigation algorithm.

Upon running the simulation, Pyrosim exports a CSV file containing temperature values at all nodes of the computational mesh. The number of temperature samples depends on the mesh resolution: finer meshes produce denser temperature data grids, enabling higher fidelity thermal representation at the cost of increased computation. The CSV output is subsequently processed by a custom ROS node, which publishes tempera-

ture data corresponding to the drone's queried position within the simulation environment.

For additional fire scenarios—such as multiple fire sources and non-symmetric temperature fields—analytical temperature maps are generated using predefined mathematical equations.

$$\frac{\Delta T_x}{\Delta T_{\text{ref}}} = f_1(N) e^{f_2(N)K_1x} \quad (1)$$

Here N is number of smoke diversion times. The parameter K_1 represents attenuation coefficient. x is distance from fire source (m). ΔT_x is maximum smoke temperature rise from the fire source ($^{\circ}\text{C}$) and ΔT_{ref} is maximum smoke temperature rise at the reference point ($^{\circ}\text{C}$). These formulations are inspired by established thermal distribution models in the fire-simulation literature [1] and are used in place of Pyrosim outputs to achieve significantly faster computation while preserving realistic temperature behavior. In these cases, secondary fire sources and heat sinks are modeled using exponential distributions, and their combined effect is obtained through the superposition principle. This approach allows controlled manipulation of thermal landscapes and enables systematic evaluation of the navigation policy under diverse and complex fire conditions.

C. Reinforcement Learning Algorithms

In our work, we have implemented the multi-arm bandit algorithm with two different approaches, i.e., the ϵ -greedy approach and the Upper Confidence Bound (UCB) approach, and compared them with the traditional hill climbing algorithm with a greedy approach. Details of the algorithms are given in the following part.

1) Hill Climbing Algorithm with Greedy approach:

The pseudo-code for the algorithm is given below.

Algorithm 1 Hill Climbing Algorithm - Greedy Approach

```

Initialize current position of the drone
Set converged  $\leftarrow$  false
while not converged do
  Go to the points,  $\{(x, y) \in \mathbb{R}^2 | x^2 + y^2 = r^2\}$ 
  Check convergence
  if convergence criteria satisfied then
    Set converged  $\leftarrow$  true
    break
  end if
  Determine direction corresponding to maximum gradient
  for  $i = 1$  to  $k$  do
    Move to the next step in direction of max gradient
    Update current position of the drone
    Check convergence
    if convergence criteria satisfied then
      Set converged  $\leftarrow$  true
      break
    end if
  end for
end while

```

In each step, while performing a 360° about the current position of the drone, the temperature gradient at each point

on the perimeter of the circle is simply calculated as the difference between the temperature at the current position and the temperature at the center of the circle.

2) *Multi-arm Bandit Algorithm with ϵ -Greedy Approach:* In Multi-arm bandit algorithms ϵ -greedy based approach is standard and widely used for localization [6]. The pseudo-code for the algorithm is given below.

Algorithm 2 Multi-Armed Bandit Algorithm - ϵ Greedy Approach

Require: Exploration probability ϵ , number of directions K

```

1: Initialize count[ $i$ ]  $\leftarrow 0$  and value[ $i$ ]  $\leftarrow 0$  for all directions  $i = 1, \dots, K$ 
2: Initial Exploration:
3: for  $i = 1$  to  $K$  do
4:    $g \leftarrow$  sample temperature gradient( $i$ )
5:   count[ $i$ ]  $\leftarrow 1$ 
6:   value[ $i$ ]  $\leftarrow g$ 
7: end for
8: while true do
9:   Generate a uniform random number  $r \in [0, 1]$ 
10:  if  $r < \epsilon$  then
11:    dir  $\leftarrow$  random direction
12:  else
13:    dir  $\leftarrow \arg \max_i \text{value}[i]$ 
14:  end if
15:  Go to the next step in the suggested direction (dir)
16:   $g \leftarrow$  Sample temperature gradient at current position
17:  count[dir]  $\leftarrow \text{count}[\text{dir}] + 1$ 
18:  Update estimate:
      
$$\text{value}[\text{dir}] \leftarrow \text{value}[\text{dir}] + \frac{g - \text{value}[\text{dir}]}{\text{count}[\text{dir}]}$$

19:  Check convergence
20:  if convergence criteria satisfied then
21:    break
22:  end if
23: end while

```

3) *Multi-arm Bandit Algorithm with UCB Approach:* In this approach, unlike ϵ -greedy, upper confidence bound for each arm is calculated using the following formula [7].

$$\text{UCB} = \text{value} + c \sqrt{\frac{2 \ln(Ts)}{\text{Count}}} \quad (2)$$

where, *value* is the empirical mean of the temperature gradient along an arm. c is the exploration constant. Ts is the total number of steps taken and *Count* is the number of times an arm is exploited.

The next step is always taken in the direction having the maximum value of the upper confidence bound.

The estimated location of the fire source is determined based on the convergence criteria. The pseudo-code for the convergence criteria used in the algorithms is given below.

Algorithm 3 Convergence Criteria

Require: Maximum number of scans $N_{\max}$, temperature threshold T_{th}

- 1: **When** convergence is checked
 - 2: **if** temperature decreases for 4 consecutive steps while moving in the suggested direction **then**
 - 3: Estimate fire source location as the position from where the first drop in temperature happened
 - 4: **if** temperature at estimated location $> T_{th}$ **then**
 - 5: convergence criteria is satisfied
 - 6: **return**
 - 7: **end if**
 - 8: **end if**
 - 9: **if** $n > N_{\max}$ **then**
 - 10: convergence criteria is satisfied
 - 11: **return**
 - 12: **end if**
-

In both of the above multi arm-bandit algorithms, the exploration constant is dynamically updated based on the variance of the number of times the arms are exploited. The pseudo-code for the update rule is given in the following.

Algorithm 4 Adaptive Update Rule of Exploration Constant

- Require:** Array of visit counts $count[i]$
- 1: Initialize ExpCons (exploration constant)
 - 2: Compute variance of visit counts:

$$var \leftarrow \text{Variance}(count[i])$$

- 3: Compute mean count with stability term:

$$k \leftarrow \text{Mean}(count[i]) + 1.0$$

- 4: Compute adaptive factor:

$$factor \leftarrow cons \times \left(1 - \frac{var}{var + k}\right), \{cons \in (0, 1]\}$$

- 5: Clip the factor:

$$factor \leftarrow \text{clip}(factor, Min, Max)$$

- 6: Update exploration constant:

$$\text{ExpCons} \leftarrow \text{ExpCons} + factor$$

D. ROS Webots and RL Framework Pipeline

The pipeline begins with Pyrosim-generated temperature distributions stored in CSV format, which are first parsed by the Pyrosim Temperature Node and converted into ROS-compatible messages. These processed temperature values are then published on a dedicated Temperature Data Topic and accessed by the RL node as part of its observation space. Using the temperature input along with the drone's current position, the RL node computes the next target position. This target is then sent to the external Webot position controller, which generates velocity commands based on the error between the current and desired positions. The RL node continuously

receives updated temperature and GPS information, allowing it to iteratively issue refined target positions for closed-loop navigation.

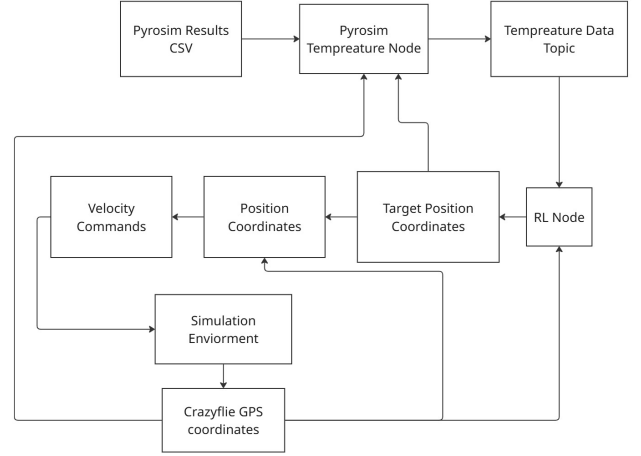


Fig. 1: Block Diagram

IV. NUMERICAL SIMULATION

We have performed simulations in a few different scenarios for three different algorithms, as already discussed above, in the following sections. The performance of the algorithms in different cases have also been compared on the basis of some key evaluation metrics, i.e., total distance traveled and mean squared error. For our experimental simulation, a 25 m. by 25 m. warehouse is considered with an environment temperature of 20 °C. The drone is launched from the global origin (0, 0) in all cases.

The hyper-parameters of the algorithms are highlighted in the following table.

TABLE I: Hyperparameters of the algorithms

Step Size	ϵ	c	Max Steps	Threshold Temp(°C)
0.3	Range(0.35, 0.45)	Range(0.30, 0.40)	1000	160

A. Temperature Distribution with a Single Fire Source

In this case, two situations have been taken. Firstly, the fire source is placed close to the drone and, secondly, it is placed far away from the starting position of the drone, where it will encounter the problem of flat gradient.

1) *Drone close to Fire Source:* A single fire source having a maximum temperature of 200°C is placed at a global position of (13.5, 13) in the Webot simulation environment. The simulation results show that **Algorithm 1** performs best in this scenario followed by **Algorithm 3**. The drone trajectories for both the algorithms along with the temperature distribution are given in the following figure.

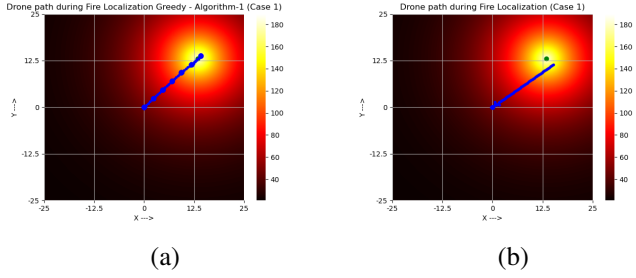


Fig. 2: (a) Hill climbing algorithm with greedy approach (b) Multi-arm bandit algorithm with UCB approach

The key performance metrics for all three algorithms are given in the following table.

TABLE II: Evaluation Metric

Metrics	Algorithm 1	Algorithm 2	Algorithm 3
Dist. Traveled (m.)	39.84	68.29	23.02
MSSE*	0.0065	0.0227	0.0170
Error (m.)	0.153	0.707	2.915

2) *Drone far away from Fire Source*: A single fire source having a maximum temperature of 200°C is placed at a global position of (23.5,23) in the Webot simulation environment. The simulation results clearly show that **Algorithm 1** does not converge and among **Algorithm 2** and **Algorithm 3**, **Algorithm 2** performs better. The drone trajectories for **Algorithm 1** and **2** along with the temperature distribution are shown in the following figure.

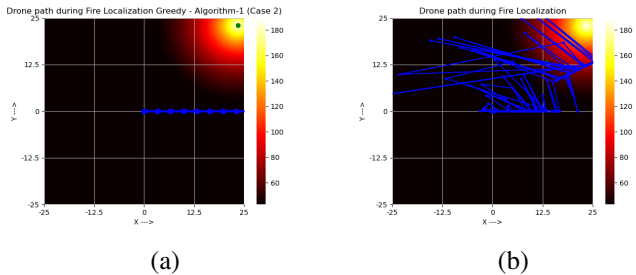


Fig. 3: (a) Hill climbing algorithm with greedy approach (b) Multi-arm bandit algorithm with ϵ -greedy approach

It is very hard to find the exact direction in case of a flat gradient, and thus leaving us no choice except bandit based algorithms, although the cost of exploration is high. The exploration is found to be less in the case of the ϵ -greedy approach compared to UCB one.

B. Temperature Distribution with a Single Primary Source and Single/Multiple Secondary Sources

In this case, we have taken a primary fire source that has a maximum temperature of 200°C at a global position of (13.5,13) along with a secondary source at a global position of (4,3.85) having a maximum temperature of 100°C . The secondary source is placed in such a way that it is on the

line joining the starting position of the drone and the primary source location and close to the Starting position of the drone. In this case, **Algorithm 2** seems to perform comparatively better than **Algorithm 3**. **Algorithm 1** as expected does not converge. The drone trajectories corresponding to the **Algorithm 1** and **2** are shown in the following figure.

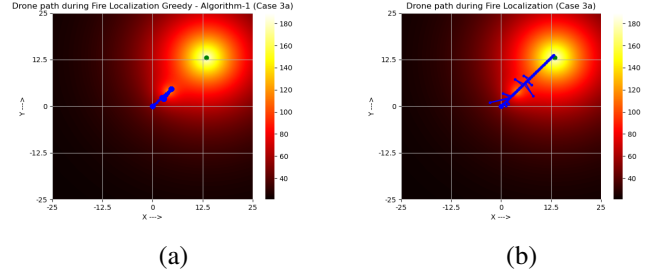


Fig. 4: (a) Hill climbing algorithm with greedy approach (b) Multi-arm bandit algorithm with ϵ -Greedy approach

The key performance metrics corresponding to all three algorithms are highlighted in the following table.

TABLE III: Evaluation Metric

Metrics	Algorithm 1	Algorithm 2	Algorithm 3
Dist. Traveled (m.)	Not Convg.	55.05	27.87
MSSE		0.0318	0.0867
Error (m.)		0.86	1.803

To study the effect of position and number of secondary sources on the performance of the Multi-arm bandite - greedy algorithm (**Algorithm 2**), we have considered the following two subcases.

1) *Effect of Position of Secondary Sources*: The secondary source is placed at three different angular positions, i.e., on the line that joins the starting position of the drone and the primary source, at an angle of 20° from the line, and at an angle of 40° from the line. The drone trajectories for 0° and 20° are given in Figures 4 and 5, respectively. The simulation results show that in the case of **Algorithm 2** with an increase in angle the drone first deviates towards the secondary source and then moves towards the primary one, but in **Algorithm 3** it is found to be more deviated in the direction of the secondary source.

2) *Effect of Multiple Secondary Sources*: Here, two different scenarios are considered. First, a single secondary source is placed at an angle of 20° from the line that joins the starting position of the drone with the primary source. Then, three secondary sources are placed along the same line. The drone trajectories for both cases are given in Figure 5.

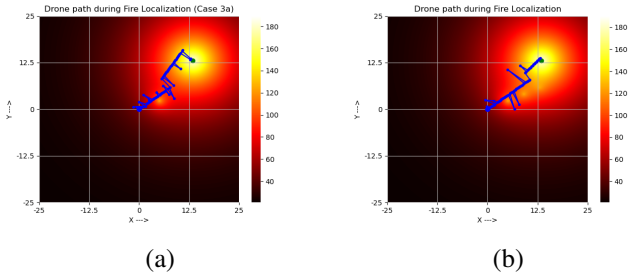


Fig. 5: Multi-arm bandit algorithm with ϵ -greedy approach in case of (a) Single secondary source at an angle of 20° (b) Multiple Secondary source at an angle of 20°

C. Temperature Distribution with a Single Primary Source and Single/Multiple Sinks

In this case, we have taken a primary fire source that has a maximum temperature of 200°C at a global position of (13.5, 13) along with a sink at a global position of (4, 3.85). The sink is placed in such a way that it is on the line that joins the starting position of the drone and the primary source location and close to the starting position of the drone. In this case, **Algorithm 1** seems to perform comparatively better than **Algorithm 2** and **3**. The drone trajectories corresponding to the **Algorithm 1** and **2** are shown in the following figure. The drone trajectory corresponding to **Algorithm 1** is shown in the following figure.

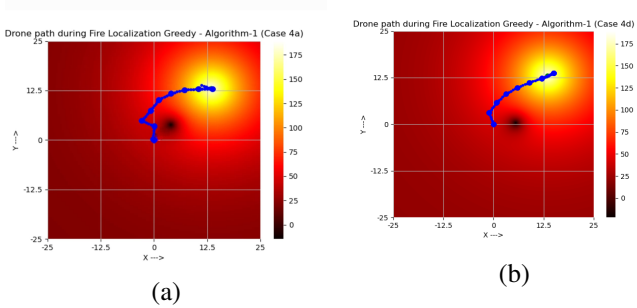


Fig. 6: Hill climbing algorithm in case of (a) Single Sink placed along the line joining primary source and starting position of the drone (b) Single Sink placed at an angle of 40°

The key performance metrics are highlighted in the following table.

TABLE IV: Evaluation Metric

Metrics	Algorithm 1	Algorithm 2	Algorithm 3
Dist. Traveled (m.)	61.52	79.97	87.64
MSSE*	0.1783	0.2741	0.2331
Error (m.)	0.2626	0.8231	1.2314

To study the effect of position on the performance of the **Algorithm 1** algorithm, we have considered the following subcase.

1) *Effect of Position of Sinks:* The sink is placed at three different angular positions, i.e., on the line that joins the starting position of the drone and the primary source, at an angle of 20° from the line, and at an angle of 40° from the line. The drone trajectories for 0° and 40° are given in Figure 6. The simulation results show that for all algorithms with decrease in angle, the deviation of the drone from the optimal path increases.

D. Control Effort

The control effort required to control the drone is found to be within range in all of the above cases. The variation of control energy has been plotted in one of such cases in the following figure.

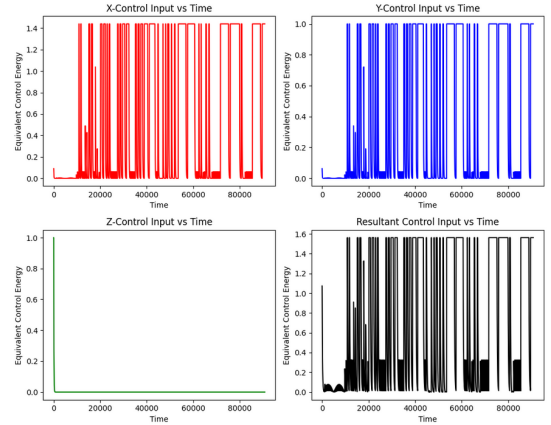


Fig. 7: Control energy plot in case of Multi-arm bandit algorithm with ϵ -greedy approach for multiple secondary sources with a single primary source

V. CONCLUSION

This work presents a complete reinforcement-learning-based navigation framework capable of guiding a micro-UAV through fire-affected environments using only temperature feedback. The performance of three different algorithms—greedy hill climbing, -greedy multi-armed bandit, and UCB bandit—was systematically evaluated across multiple thermal configurations, revealing distinct strengths and limitations for each approach.

In the single fire source case with steep gradients (fire source near the starting location), the hill-climbing algorithm consistently achieved the best performance, converging with the least distance traveled and the smallest tracking error. The strong and consistent gradient field ensured that a simple greedy ascent toward increasing temperature was reliable and efficient. In contrast, the -greedy and UCB approaches introduced additional exploratory steps, resulting in longer trajectories without significant improvement in accuracy for this scenario.

When the drone began far away from the fire source, the gradient field near the starting position was nearly flat, making greedy hill climbing ineffective; it failed to converge due to lack of directional cues. In this setting, exploration became

essential. The ϵ -greedy bandit clearly outperformed both hill climbing and UCB by maintaining a balanced exploration strategy that allowed the drone to detect the weak gradient and move toward the fire source. The UCB method showed moderate performance, but exhibited excessive early exploration, leading to longer drift before identifying the correct thermal direction.

In the case of a primary fire source with secondary fire sources, the navigation problem became significantly harder due to misleading high-temperature pockets near the starting region. The greedy hill-climbing algorithm was repeatedly trapped by these local maxima and did not converge. The ϵ -greedy bandit successfully detected and escaped these misleading peaks, demonstrating superior robustness in cluttered thermal landscapes. The UCB method reached the primary source but suffered from a greater deviation as a result of aggressive exploration, especially when the secondary sources were placed at larger angular offsets.

When thermal sinks were introduced on or near the optimal path, the temperature field exhibited sharp drops that could mislead a purely exploratory strategy. Here, hill climbing again showed the strongest performance because it aggressively followed positive gradients while avoiding regions with decreasing temperature. Both bandit algorithms were more susceptible to being drawn toward these cooler pockets, especially when the sink was placed at small angular offsets from the direct path.

Across all experiments, the bandit-based strategies demonstrated clear advantages in flat-gradient or multi-peak scenarios, while greedy hill climbing remained the most effective in clean, unimodal temperature fields or fields distorted by sinks. These consistent case-wise behaviors highlight that no single RL strategy is universally optimal, and that fire-environment navigation benefits from algorithm selection based on expected thermal complexity.

Overall, the study demonstrates that lightweight micro-UAVs can navigate hazardous thermal fields by relying solely on temperature feedback, without the need for vision or LiDAR. The proposed framework—integrating Pyrosim fire simulations, analytical thermal models, Webot, ROS, and RL—provides a strong foundation for future real-world implementations. Future work may extend this framework to dynamic fire propagation, onboard thermal sensor noise modeling, adaptive step-size control, and coordinated search using multiple UAVs.

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